## Suggested Solutions to Supplementary Exercises on Chapter 7 Vectors

**1.** **[ACJC23/Promo/Q6]**

Referred to the origin $O$, the points $P$ and $Q$ have position vectors $\mathbf{p}$ and $\mathbf{q}$ respectively, where vector $\mathbf{p}$ and vector $\mathbf{q}$ are not parallel. Point $A$ lies between $P$ and $Q$, such that $PA:AQ=\lambda:(1-\lambda)$, where $0<\lambda<1$.

**(i)** Express $\overrightarrow{OA}$ in terms of $\mathbf{p}$, $\mathbf{q}$ and $\lambda$. [1]

**(ii)** The point $B$ lies between $O$ and $Q$, such that $OB:OQ=1:3$. Given that $\overrightarrow{AB}$ is parallel to $(\mathbf{q}-4\mathbf{p})$, find the value of $\lambda$. [4]

**(iii)** It is now given that $\mathbf{p}$ is a unit vector which is perpendicular to $\mathbf{q}$ and $|\mathbf{q}|=2$, find the area of triangle $OAQ$, in terms of $\lambda$. [4]

<table>
<tr><td>1(i)</td><td>Using Ratio Theorem,<br>$\overrightarrow{OA}=\dfrac{\lambda\mathbf{q}+(1-\lambda)\mathbf{p}}{\lambda+(1-\lambda)}$<br>$=\lambda\mathbf{q}+(1-\lambda)\mathbf{p}$</td><td></td></tr>
<tr><td>1(ii)</td><td>$\overrightarrow{OB}=\dfrac{1}{3}\mathbf{q}$<br>$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}$<br>$=\dfrac{1}{3}\mathbf{q}-\left[\lambda\mathbf{q}+(1-\lambda)\mathbf{p}\right]=\dfrac{1}{3}\mathbf{q}-\lambda\mathbf{q}-(1-\lambda)\mathbf{p}$<br>$=\left(\dfrac{1}{3}-\lambda\right)\mathbf{q}+(\lambda-1)\mathbf{p}$<br>Since $\overrightarrow{AB}$ is parallel to $(\mathbf{q}-4\mathbf{p})$,<br>$\therefore \overrightarrow{AB}=m(\mathbf{q}-4\mathbf{p})$, where $m$ is a scalar.<br>$\left(\dfrac{1}{3}-\lambda\right)\mathbf{q}+(\lambda-1)\mathbf{p}=m(\mathbf{q}-4\mathbf{p})$<br>$\left(\dfrac{1}{3}-\lambda\right)\mathbf{q}+(\lambda-1)\mathbf{p}=m\mathbf{q}-4m\mathbf{p}$<br>Since vector $\mathbf{p}$ and vector $\mathbf{q}$ are not parallel,<br>by comparison,<br>$\dfrac{1}{3}-\lambda=m$ ----------(1)<br>$\lambda-1=-4m$ ----------(2)<br>Solving (1) & (2): $\lambda=\dfrac{1}{9}$.</td><td></td></tr>
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| **1(iii)** | **Area of** $\Delta OAQ = \frac{1}{2}\left\|\overrightarrow{OA}\times\overrightarrow{OQ}\right\|$<br>$= \frac{1}{2}\left\|\mathbf{a}\times\mathbf{q}\right\|$<br>$= \frac{1}{2}\left\|\left[\lambda\mathbf{q}+(1-\lambda)\mathbf{p}\right]\times\mathbf{q}\right\|$<br>$= \frac{1}{2}\left\|\lambda\mathbf{q}\times\mathbf{q} + (1-\lambda)\mathbf{p}\times\mathbf{q}\right\|$<br>$= \frac{1}{2}\left\|\mathbf{0} + (1-\lambda)\|\mathbf{p}\|\|\mathbf{q}\|\sin 90^\circ\right\|$<br>$= \frac{1}{2}\left\|(1-\lambda)(1)(2)(1)\right\|$<br>$= \|1-\lambda\|$<br>$= 1-\lambda.$ **(since** $0<\lambda<1$**)** | |

2. **[EJC23/Promo/Q4]**
Referred to the origin $O$, let $P$, $Q$ and $R$ be distinct points with position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively.

**(a)** Show that $(\mathbf{r}-\mathbf{p})\times(\mathbf{r}-\mathbf{q}) = \mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}$. [2]

**(b)** Give the geometrical meaning of $\frac{1}{2}\left|\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}\right|$. [2]

**(c)** Given that $\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}=\mathbf{0}$, $PR = 3QR$, and that $PQ > PR$, express $\mathbf{r}$ in terms of $\mathbf{p}$ and $\mathbf{q}$. [3]

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| **2(a)** | [Consider any triplet of distinct points $P$, $Q$, $R$ with position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively.]<br>LHS $= (\mathbf{r}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})$<br>$= (\mathbf{r}-\mathbf{p})\times\mathbf{r}-(\mathbf{r}-\mathbf{p})\times\mathbf{q}$<br>$= \mathbf{r}\times\mathbf{r}-\mathbf{p}\times\mathbf{r}-\mathbf{r}\times\mathbf{q}+\mathbf{p}\times\mathbf{q}$<br>$= \mathbf{r}\times\mathbf{p}+\mathbf{q}\times\mathbf{r}+\mathbf{p}\times\mathbf{q}$ $(\because \mathbf{r}\times\mathbf{r}=0 \text{ and } \mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a})$<br>$= \mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}$<br>$=$ RHS<br>$\therefore\ \mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}=(\mathbf{r}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})$ (shown) | |
| **2(b)** | $\frac{1}{2}\left\|\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}\right\| = \frac{1}{2}\left\|(\mathbf{r}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\right\|$<br>$= \frac{1}{2}\left\|\left(\overrightarrow{OR}-\overrightarrow{OP}\right)\times\left(\overrightarrow{OR}-\overrightarrow{OQ}\right)\right\|$<br>$= \frac{1}{2}\left\|\overrightarrow{PR}\times\overrightarrow{QR}\right\|$<br>$\therefore\ \frac{1}{2}\left\|\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}\right\|$ represents the area of $\Delta PQR$. | |

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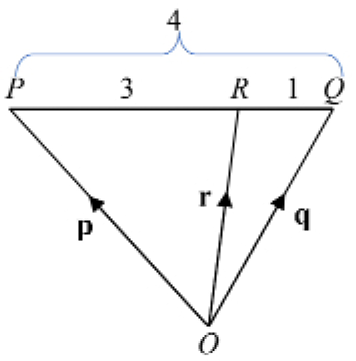
| | Alternative<br>$\frac{1}{2}\lvert\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}\rvert$ represents half the area of a parallelogram with $PR$ and $QR$ as adjacent sides. | |
| **2(c)** | Given $\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}=\mathbf{0}$ ,<br>$\Rightarrow\ (\mathbf{r}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})=\mathbf{0}$ (from result in **(a)**)<br>$\Rightarrow\ \overrightarrow{PR}\times\overrightarrow{QR}=\mathbf{0}$<br>$\because\ P,Q,R$ are distinct points,<br>$\Rightarrow\ \overrightarrow{PR}\neq\mathbf{0}$, and $\overrightarrow{QR}\neq\mathbf{0}$ ,<br>$\Rightarrow\ \overrightarrow{PR}\parallel\overrightarrow{QR}$ , i.e. $P,Q,R$ are collinear points.<br>Given also that $PR=3QR$, and $PQ>PR$,<br>$\therefore$ Point R divides PQ internally in the ratio 3:1, i.e.<br>$PR:RQ=3:1$.<br><br>By the ratio theorem, position vector $\mathbf{r}=\frac{\mathbf{p}+3\mathbf{q}}{4}$ .<br>Alternative (to show collinear)<br>Given $\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}=\mathbf{0}$ ,<br>$\Rightarrow$ Area of $\Delta PQR=\frac{1}{2}\lvert\mathbf{p}\times\mathbf{q}+\mathbf{q}\times\mathbf{r}+\mathbf{r}\times\mathbf{p}\rvert$ (by part **(b)**)<br>$=\frac{1}{2}\lvert\mathbf{0}\rvert$<br>$=0$ i.e. $\Delta PQR$ is a degenerate triangle<br>i.e. $P,Q,R$ are collinear points. | |

**3.** **[SAJC22/Promo/Q9]**

Two distinct points $A$ and $B$ have non-zero position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively.
The point $R$, with position vector $\mathbf{r}$, varies such that $\mathbf{r}\times\mathbf{a}=\mathbf{b}\times\mathbf{a}$.

**(i)** Show that $\mathbf{r}-\mathbf{b}=k\mathbf{a}$ where $k$ is a real number. [2]

**(ii)** Describe geometrically the set of all possible positions of the point $R$. [2]

**(iii)** Given further that $\angle AOB=90^\circ$, show that $\lvert\mathbf{r}.\hat{\mathbf{b}}\rvert=\lvert\mathbf{b}\rvert$. [4]